

Statistics Lecture 5

Hypothesis testing pt. 1

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AALBORG UNIVERSITY
DENMARK

Connectivity

Schedule

Lecture

1. Introduction to statistics
2. Parameter estimation
3. Confidence intervals
4. Regression

5. Hypothesis testing pt. 1

6. Hypothesis testing pt. 2

Workshop: Wrap-up and exam problems

When

March 12

March 24

March 26

March 31

April 7

April 9

April 28?

Outline

1. **Recap on confidence intervals**
2. **Definition and classification of tests**
3. **Tests with normal population and known variance**
4. **Hypothesis testing for Bernoulli population**

Sanity check

- What is a confidence interval?

Range of values between two numbers that contain the true value of the parameter we are estimating with some confidence level

- What types of confidence intervals are there?

One-sided (right-handed and left-handed) and two-sided

- How do we calculate α from the confidence level?

Confidence level is $1 - \alpha$

$$\text{confidence} = (1 - \alpha)100 \quad \alpha = 1 - \frac{\text{confidence}}{100}$$

- What is better: a wide or a narrow confidence interval?

1. Recap on confidence intervals

Confidence intervals

Our estimators are RVs, so they must have a distribution

The confidence interval (CI) is the best answer to the question:

What is the range of values $C_{1-\alpha} = (a, b)$ around the estimate $\hat{\theta}_n$ such that we are confident with probability $1 - \alpha$ that the true value θ is inside the range?

$$P(\theta \in C_{1-\alpha}) \geq 1 - \alpha$$

Usually $\alpha = 0.05$ so we look at the 95% confidence interval

$$C_{0.95} = (a, b)$$

Confidence intervals for normal RVs with known σ^2

If $X_1, X_2, \dots, X_n \sim N(\mu, \sigma^2)$, we know that the MLE of the μ is the sample mean

Not known $\hat{\mu}_n = \frac{1}{n} \sum_{i=1}^n X_i \sim N\left(\mu, \frac{\sigma^2}{n}\right)$ known

Therefore, the variance of our estimator is $\text{var}(\hat{\mu}_n) = \sigma^2/n$ and the CI is

$C_{1-\alpha} = \left(\hat{\mu}_n - Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \hat{\mu}_n + Z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right)$ std. deviation

quantile

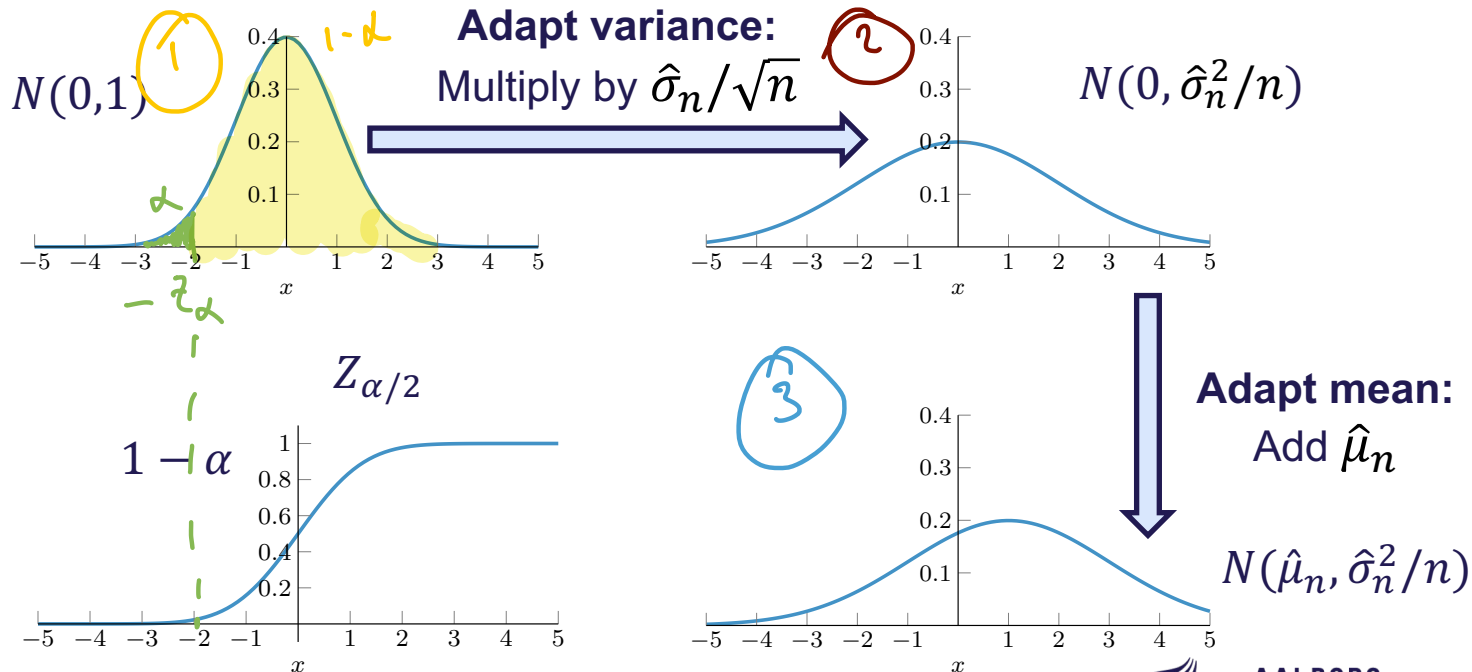
For normal RVs, 95% of outcomes are between $\hat{\mu}_n - 1.96\sigma$ and $\hat{\mu}_n + 1.96\sigma$

$C_{0.95} = \left(\hat{\mu}_n - 1.96 \frac{\sigma}{\sqrt{n}}, \hat{\mu}_n + 1.96 \frac{\sigma}{\sqrt{n}} \right)$ std. Normal

Why this formula?

$$C_{1-\alpha} = \left(\hat{\mu}_n - Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \hat{\mu}_n + Z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right)$$

So we don't have to obtain the quantiles $1 - \alpha$ for each distribution $N(\hat{\mu}_n, \hat{\sigma}_n^2/n)$



Two options for calculating CIs with known variance

Step 1: Find the value of your estimate $\hat{\mu}_n$ and its standard deviation $\frac{\sigma}{\sqrt{n}}$

Step 2: Choose one:

1. Using $N(0,1)$: Find $Z_{\alpha/2}$ and, then, $\hat{\mu}_n \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$

Matlab: norminv(1 - $\alpha/2$, 0, 1)

Python: scipy.stats.norm.ppf(1 - $\alpha/2$)

2. Using $N(\hat{\mu}_n, \frac{\sigma^2}{n})$: Find the quantiles directly

Matlab: norminv(1 - $\alpha/2$, $\hat{\mu}_n$, $\frac{\sigma}{\sqrt{n}}$)

Python: scipy.stats.norm($\hat{\mu}_n$, $\frac{\sigma}{\sqrt{n}}$).ppf(1 - $\alpha/2$)

The CI cheat sheet for Gaussian RVs

Known σ^2 ?	Type of CI	Quantile	CI formula $C_{1-\alpha}$
Yes	Two-sided	$Z_{\alpha/2}$	$(\hat{\mu}_n - Z_{\alpha/2}\sigma/\sqrt{n}, \hat{\mu}_n + Z_{\alpha/2}\sigma/\sqrt{n})$
Yes	Right-handed	Z_{α}	$(\hat{\mu}_n - Z_{\alpha}\sigma/\sqrt{n}, \infty)$
Yes	Left-handed	Z_{α}	$(-\infty, \hat{\mu}_n + Z_{\alpha}\sigma/\sqrt{n})$
No	Two-sided	$T_{n-1,\alpha/2}$	$(\hat{\mu}_n - T_{n-1,\alpha/2}\hat{\sigma}_n/\sqrt{n}, \hat{\mu}_n + T_{n-1,\alpha/2}\hat{\sigma}_n/\sqrt{n})$
No	Right-handed	$T_{n-1,\alpha}$	$(\hat{\mu}_n - T_{n-1,\alpha}\hat{\sigma}_n/\sqrt{n}, \infty)$
No	Left-handed	$T_{n-1,\alpha}$	$(-\infty, \hat{\mu}_n + T_{n-1,\alpha}\hat{\sigma}_n/\sqrt{n})$

* Use t-distribution for all cases without Gaussian RVs

2. Definition and classification of tests

Types of tests based on the populations

Parameter estimation with 1 population

There is some **idea** about the value of a parameter of F

Is the **idea** correct?

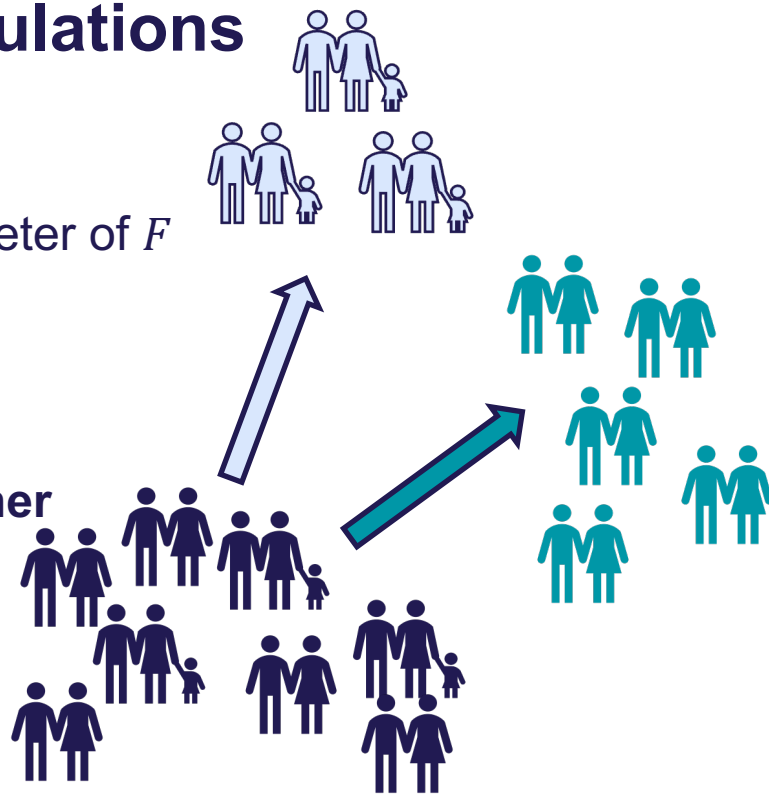
Example: Is it true that the average age is 20?

Compare multiple populations with each other

No parameter known a priori

- Begin with different populations
- One initial population divided into groups

Can we find differences between populations?



How do we conduct the tests?

We begin with two opposing hypotheses:

H_0 : The **null hypothesis**, the one assumed to be true

H_1 : The **alternative hypothesis**, which contradicts H_0

We try to find evidence to support H_0

If we cannot, then we say we can **reject** H_0

Accepting a hypothesis does not mean it is true, but that the data support it

Example: Which should be the null hypothesis in a pharmaceutical trial?

a) The new drug is effective

b) The new drug is **not** effective ★

How do we formulate the hypotheses?

H_0 : The null hypothesis is the baseline case with a simple formulation

Example: A parameter θ takes values in the set Θ and we test the value of θ

$$\mathbf{H}_0: \theta = \theta_0 \in \Theta$$

$$\mathbf{H}_1: \theta \neq \theta_0$$

If we reject H_0 , then it means that $\mathbf{H}_1: \theta \neq \theta_0$ is true but still $\theta \in \Theta$

We could also define the hypotheses as $\mathbf{H}_0: \theta \in \Theta_0$ versus $\mathbf{H}_1: \theta \notin \Theta_0$

This implies $\theta \in \Theta_1$ and $\Theta_0 \cup \Theta_1 = \Theta$

Types of tests with a single population

One-sided tests

Null hypothesis: The true value lies in an interval with one finite limit

$H_0: \theta \in \Theta$ where $\Theta = (-\infty, b]$ or $\Theta = [a, \infty)$

The same as $H_0: \theta \leq b$ or $\theta \geq a$

Two-sided tests

Null hypothesis: the true value lies in an interval with finite limits

$H_0: \theta \in \Theta = [a, b]$

The same as $H_0: a \leq \theta \leq b$

There is a special case: If Θ has a single element $\theta = a = b$

Procedure for testing

1. Choose a parameter for testing θ
2. Formulate the null hypothesis H_0 and the alternative hypothesis H_1 about θ
3. Design the test (test statistic or p-value) and define the rejection region R
4. Collect a sample $X_1, X_2, X_3, \dots, X_n$ of i.i.d. RVs
5. If the observation regarding θ is **close enough** to the value(s) in H_0

H_0 cannot be rejected

6. Else:

Reject H_0 and accept H_1

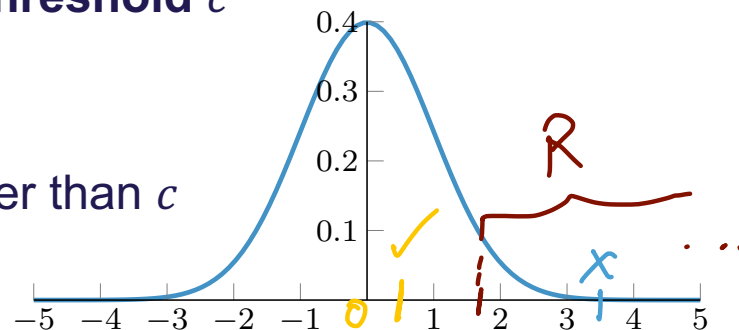
Two ways of testing

1. Get the test statistic T and compare it to a threshold c

These values are points in the x-axis

Define rejection region $R = \{x: T(x) > c\}$

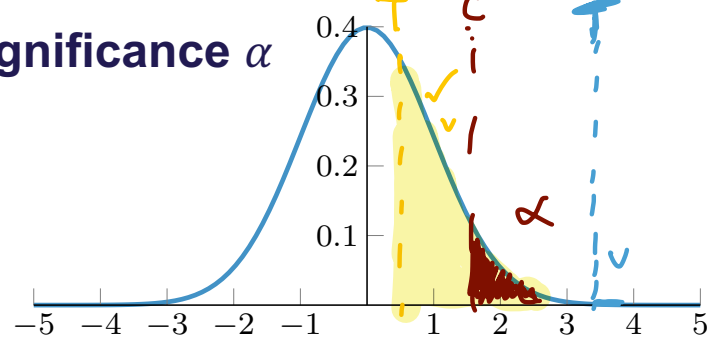
Reject the null hypothesis if T is farther from center than c



2. Comparing the p-value v with the level of significance α

These are the areas under the curve

Reject the null hypothesis if $v < \alpha$



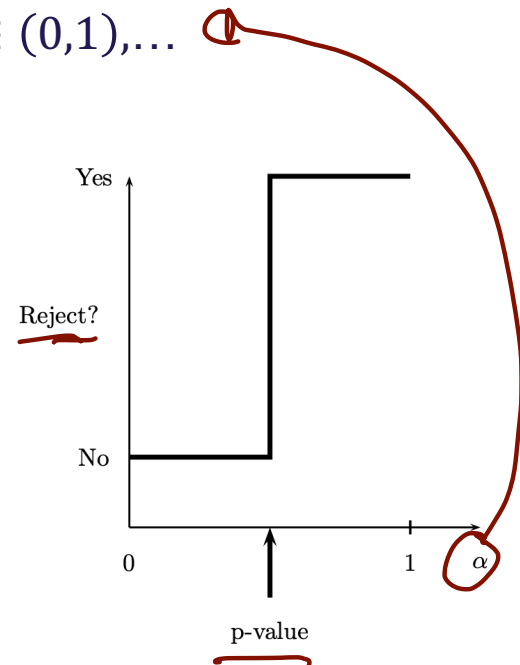
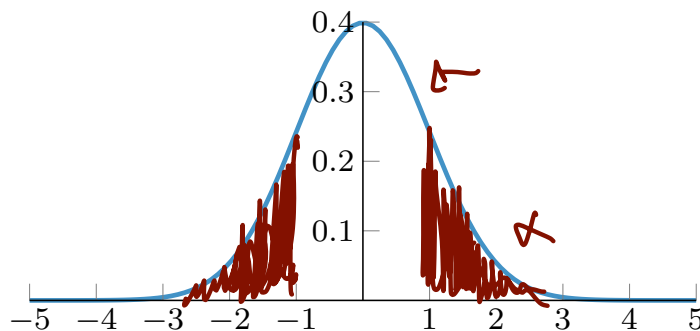
The p-value

If a test rejects the null hypothesis at significance level $\alpha \in (0,1), \dots$
...then, it will also reject the test for any $\alpha' > \alpha$

We can calculate the smallest α at which a test is rejected

Definition:

The p-value is the smallest level at which we can reject H_0



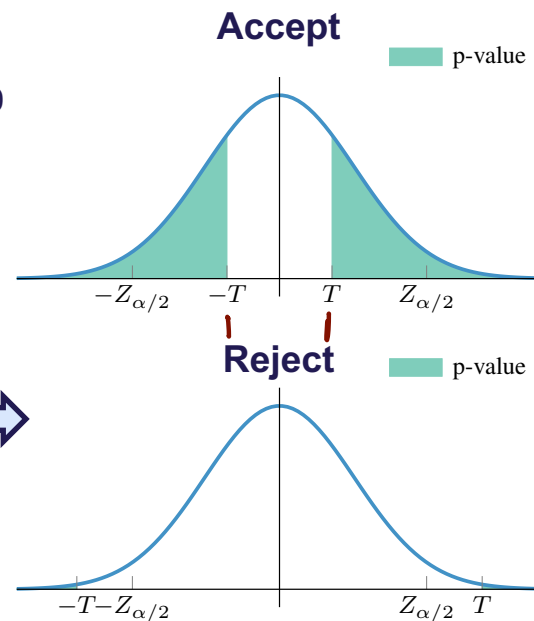
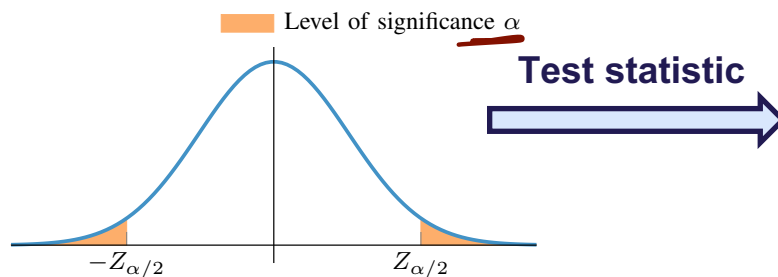
Test statistic and p-value

H_0 : The **null hypothesis**, the one assumed to be true

H_1 : The **alternative hypothesis**, which contradicts H_0

We try to find evidence to reject H_0

But we don't want to reject H_0 when it's true: α



What if we want to find relations between populations?

The p-value: interpretation

The p-value is the **probability** (under H_0) of observing a value of the test statistic the same as or more extreme than what was actually observed

The p-value for a two-sided test with test statistic T and RV $Z \sim N(0,1)$ is

$$v = P(|Z| > T) = 2P(Z > T) = 2(1 - \Phi(T))$$

Test outcomes and errors

Errors can occur even when we follow a proven methodology

If a test statistic is in the rejection region R , H_0 is rejected

	Accept	Reject
H_0 is True	All good!	Type I error
H_0 is False	Type II error	All good!

False negative (next to Type I error)

False positive (below Type II error)

Significance level α

The probability of Type I error should not exceed α

$$P(\text{Type I error}) = \alpha$$

We want α to be small: We don't want to reject H_0 when it's true

Trade-off between Type I and Type II errors

We have to evaluate the consequences of Type I and Type II errors

If we reduce α , we have fewer Type I errors, but more Type II errors

The choice of α depends on the application:

Legal system: Innocent (H_0) until proven guilty

Computer security: H_0 is that the user fingerprint is incorrect (unlock the phone)

Medicine: H_0 is the patient healthy or sick?

Exercise: Is it a fair coin? Pt. 1

Before you toss a coin to “make” an important decision, you want to make sure that the coin is fair. How do you do it?

Procedure

1. Choose a parameter for testing θ

outcomes $w \in \{0, 1\}$

Each coin toss is a RV $X_i \sim \text{Bernoulli}(p)$

So $\theta = p$

2. Formulate the null hypothesis H_0 and the alternative hypothesis H_1

A fair coin should have $\theta = p_0 = 0.5$

a) The coin is fair if $p = 0.5$ H_0

b) The coin is **not** fair if $p \neq 0.5$ H_1

Exercise: Is it a fair coin? Pt. 2

3. Design the test and define the rejection region

$H_0: \theta = 0.5$

Toss the coin n times and count the number of heads

Generate an estimate for θ as

$$\hat{\theta}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

$$T = |\hat{\theta}_n - 0.5|$$

We define our **test statistic** as $T = |\hat{\theta}_n - \theta_0|$ where $\theta_0 = p_0 = 0.5$

If $T = |\hat{\theta}_n - \theta_0| \leq c$, we cannot reject H_0

Otherwise, if $T = |\hat{\theta}_n - \theta_0| > c$, we reject H_0

Rejection region:

$$\alpha = 0.05$$
$$R = \{x : T > c\}$$

Exercise: Is it a fair coin? Pt. 3

4. Collect a sample $X_1, X_2, X_3, \dots, X_n$ of i.i.d. RVs

5. If the observation regarding θ is close enough to the value(s) in H_0

H_0 cannot be rejected

6. Else:

Reject H_0 and accept H_1

How do we choose c and α ?

We're not done yet

We still need to define the probability of Type I error α

$$P(\text{Type I error}) = P(T > c \mid H_0 \text{ is True}) = \alpha$$

This means that the probability that we fall in the rejection region given that H_0 is True should be, at most, equal to α

How do we choose c and α ?

We know that

$$\hat{p}_n = \frac{1}{n} \sum_{i=1}^n X_i \approx N\left(p, \frac{\sigma^2}{n}\right)$$

And we have a powerful definition of **Confidence interval (CI)**:

Range of values $C_{1-\alpha} = (a, b)$ around the estimate $\hat{\theta}_n$ such that:

we are confident with probability $1 - \alpha$ that the **true value** θ is inside the range

$$P(\theta \in C_{1-\alpha}) \geq 1 - \alpha$$

So, we can use what we know about CIs for testing, at least in principle

3. Tests with normal population and known variance

Two-sided test for the mean

Two-sided test for μ with normal population and known σ

The null hypothesis $H_0: \mu = \mu_0$, so $H_1: \mu \neq \mu_0$

Test statistic: $T = |\hat{\mu}_n - \mu_0| \sqrt{n} / \sigma$

Rejection region: $R = \{X_1, X_2, \dots, X_n : T > c\}$

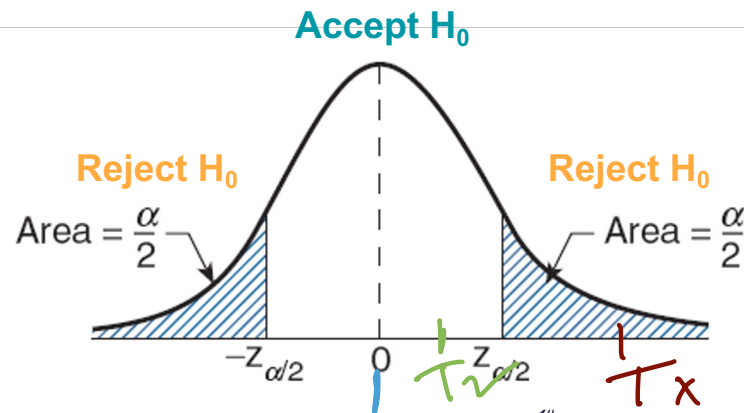
The total area of the region outside the CI is α

$$C_{1-\alpha} = \left(\hat{\mu}_n - Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \hat{\mu}_n + Z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right)$$

So $c = Z_{\alpha/2}$ In CIs, $1-\alpha$ confidence
In tests, α is significance

Reject H_0 if $\frac{|\hat{\mu}_n - \mu_0| \sqrt{n}}{\sigma} > Z_{\alpha/2}$

confidence $= (1 - \alpha) / 100$



Example

A signal of value μ is sent from location A to location B.

Gaussian noise is added to the signal is a RV with distribution $N(0,4)$.

Then, the received signal is normally distributed with mean μ and $\sigma = 2$.

It is suspected that $\mu = 8$. Test this hypothesis when the signal value is sent 5 times and the sample average at location B is $\bar{X}_5 = 9.5$.

Use a 5% level of significance. $\alpha = 0.05$

$$X_i = \mu + \epsilon_i$$

1. Define the hypothesis
2. Compute the test statistic
3. Compute the rejection region
4. Decide on acceptance

Solution

We know that the MLE for parameter μ is $\hat{\mu}_5 = \bar{X}_5 = 9.5$

1. $H_0: \mu_0 = 8$

2. We accept H_0 if $\frac{|\hat{\mu}_n - \mu_0| \sqrt{n}}{\sigma} \leq Z_{\alpha/2}$. Therefore, the test statistic is $T = \frac{|\hat{\mu}_n - \mu_0| \sqrt{n}}{\sigma}$

3. We recall that, for $\alpha = 0.05$, we have $Z_{\alpha/2} = 1.96$

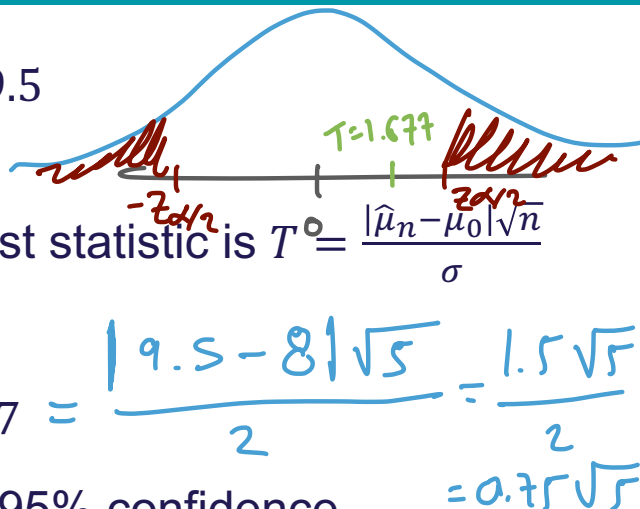
4. We cannot reject H_0 because $T = \frac{|\hat{\mu}_n - \mu_0| \sqrt{n}}{\sigma} = 1.677 = \frac{|9.5 - 8| \sqrt{5}}{2} = \frac{1.5 \sqrt{5}}{2} = 0.75 \sqrt{5}$

By doing this, our CI includes the real value of μ with 95% confidence

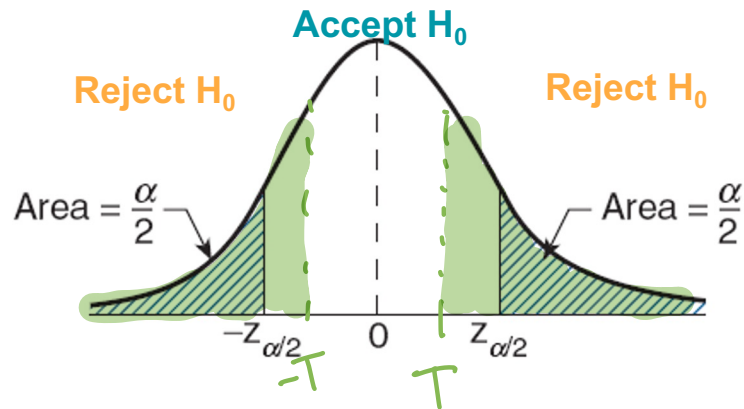
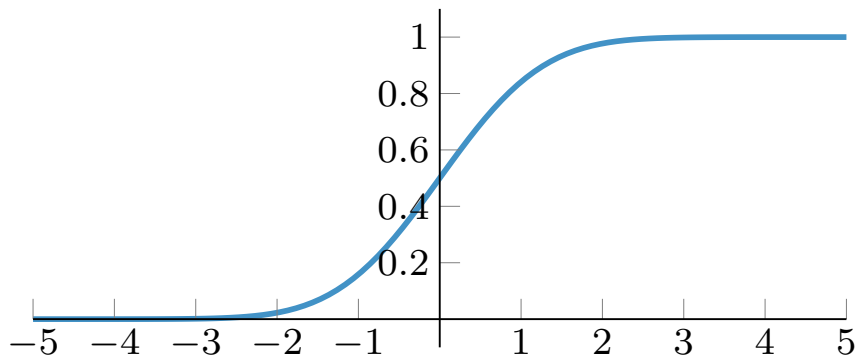
$T = 1.677$ and $Z_{\alpha/2} = 1.96$

How close was the decision to reject or not?

Accept H_0 because
 $T = 1.677 \leq Z_{\alpha/2} = 1.96$



How close was the test?



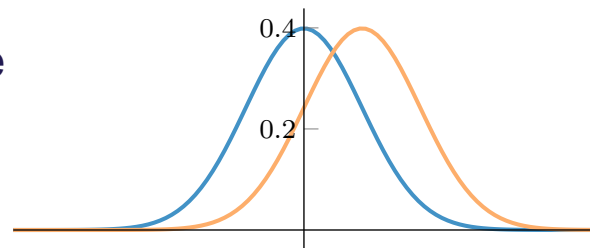
Type II error for two-sided test

The probability of Type II error is the probability of not rejecting H_0 given its False

$$P(\text{Type II error}) = P(T < c \mid H_0 \text{ is False}) = \beta$$

For our case, H_0 is false if $\mu \neq \mu_0$ and we can calculate

$$\begin{aligned}\beta(\mu) &= P\left(\frac{|\hat{\mu}_n - \mu_0|\sqrt{n}}{\sigma} < Z_{\alpha/2} \mid H_0 \text{ is False}\right) \\ &= P\left(\mu_0 - Z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \leq \hat{\mu}_n \leq \mu_0 + Z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \mid H_0 \text{ is False}\right) \\ &= \Phi\left(\frac{(\mu_0 - \mu)\sqrt{n}}{\sigma} + Z_{\alpha/2}\right) - \Phi\left(\frac{(\mu_0 - \mu)\sqrt{n}}{\sigma} - Z_{\alpha/2}\right)\end{aligned}$$



One-sided test for the mean

One-sided test for μ with normal population and known σ

The null hypothesis $H_0: \mu \leq \mu_0$, so $H_1: \mu > \mu_0$

Test statistic: $T = (\hat{\mu}_n - \mu_0)\sqrt{n}/\sigma$

Rejection region: $R = \{X_1, X_2, \dots, X_n : T > c\}$

The total area of the region outside the CI is α

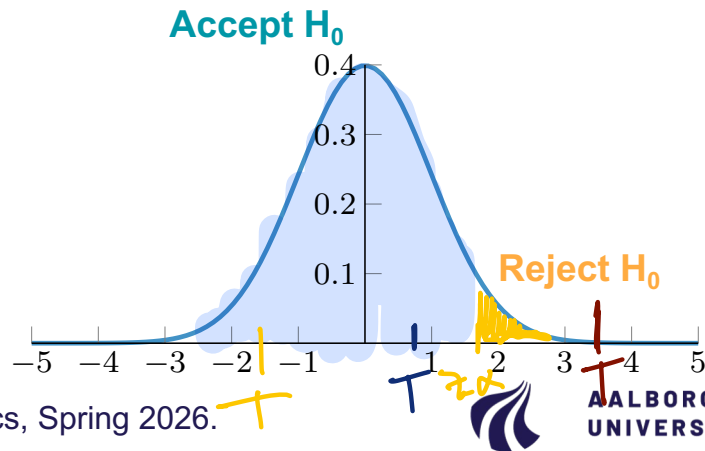
$$C_{1-\alpha} = \left(-\infty, \hat{\mu}_n + Z_\alpha \frac{\sigma}{\sqrt{n}}\right)$$

So $c = Z_\alpha$

Reject H_0 if $\frac{(\hat{\mu}_n - \mu_0)\sqrt{n}}{\sigma} > Z_\alpha$

if $\hat{\mu}_n - \mu_0 \leq 0$

if $\hat{\mu}_n - \mu_0 \geq 0$



One-sided test for μ with normal population and known σ

The null hypothesis is $H_0: \mu \geq \mu_0$, so $H_1: \mu < \mu_0$

Test statistic: $T = (\hat{\mu}_n - \mu_0)\sqrt{n}/\sigma$

Rejection region: $R = \{X_1, X_2, \dots, X_n : T < c\}$

Changed!

Same as before

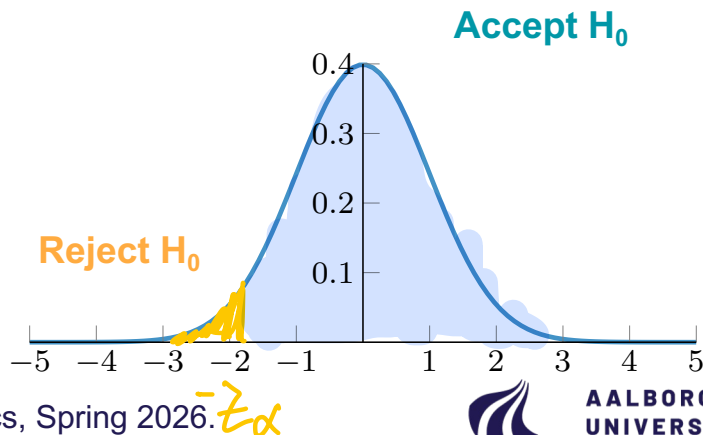
Changed!

The total area of the region outside the CI is α

$$C_{1-\alpha} = \left(\hat{\mu}_n - Z_\alpha \frac{\sigma}{\sqrt{n}}, \infty \right)$$

So $c = -Z_\alpha$

Reject H_0 if $\frac{(\hat{\mu}_n - \mu_0)\sqrt{n}}{\sigma} < -Z_\alpha$



Example

Go back to the previous example of the transmitted signal.

Assume that we want to test whether the signal is 8 or less with $\bar{X}_5 = 9.5$.

What is the conclusion?

1. Define H_0
2. Can we reject H_0 ?
3. What is the p-value?

The p-value for a one-sided test with test statistic T and RV $Z \sim N(0,1)$ is

$$v = \underline{P(Z > T)} = 1 - \Phi(T)$$

Solution

We know that $\hat{\mu}_5 = \bar{X}_5 = 9.5$

1. Now $H_0: \mu_0 \leq 8$ and $H_1: \mu_1 > 8$
2. We get $T = \frac{(\hat{\mu}_n - \mu_0)\sqrt{n}}{\sigma} = 1.677 > Z_\alpha = 1.645$, so **we reject** H_0
3. The p-value is $v = 1 - \Phi(1.677) = 0.0467 < \alpha = 0.05$

Type II error for one-sided test

The probability of Type II error is the probability of not rejecting H_0 given its False

$$P(\text{Type II error}) = P(T < c \mid H_0 \text{ is False}) = \beta$$

For our case, H_0 is false if $\mu > \mu_0$ and we can calculate

$$\begin{aligned}\beta(\mu) &= P\left(\hat{\mu}_n - \mu_0 < -Z_\alpha \frac{\sigma}{\sqrt{n}} \mid H_0 \text{ is False}\right) \\ &= \Phi\left(\frac{(\mu_0 - \mu)\sqrt{n}}{\sigma} - Z_{\alpha/2}\right)\end{aligned}$$

Summary table

Testing for the mean of one normal population with known variance σ

H_0	H_1	Test statistic T	Rejection region	P-value
$\mu = \mu_0$	$\mu \neq \mu_0$	$ \hat{\mu}_n - \mu_0 \sqrt{n} / \sigma$	$T > Z_{\alpha/2}$	$2(1 - \Phi(T))$
$\mu \leq \mu_0$	$\mu > \mu_0$	$(\hat{\mu}_n - \mu_0) \sqrt{n} / \sigma$	$T > Z_{\alpha}$	$(1 - \Phi(T))$
$\mu \geq \mu_0$	$\mu < \mu_0$	$(\hat{\mu}_n - \mu_0) \sqrt{n} / \sigma$	$T < -Z_{\alpha}$	$\Phi(T)$

CDF of $N(0,1)$

Two-sided
one-sided

$$T = \frac{|\hat{\mu}_n - \mu_0| \sqrt{n}}{\sigma} > Z_{\alpha/2}$$

$$CI = \left(\hat{\mu}_n - Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \hat{\mu}_n + Z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right)$$

$$|\hat{\mu}_n - \mu_0| > Z_{\alpha/2} \sigma / \sqrt{n}$$

4. Hypothesis testing for Bernoulli population

Bernoulli RVs

Bernoulli RVs have only two possible outcomes: 1 or 0

$$P(X_i = 1) = p \text{ and } P(X_i = 0) = 1 - p$$

The sum of $X_1, X_2, \dots, X_n \sim \text{Bernoulli}(p)$ RVs is a binomial RV with parameters n, p :

$$X = \sum_{i=1}^n X_i \sim B(n, p)$$

where $P(X = k; n, p) = \binom{n}{k} p^k (1 - p)^{n-k}$ and $\binom{n}{k} = \frac{n!}{k!(n-k)!}$

We also know that the MLE for p is

$$\hat{p}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

One-side tests with p-value for Bernoulli RVs

We formulate a null hypothesis $H_0: p \leq p_0$

We begin with an observation where k out of the n outcomes are 1, so $X = k$

Recall: The p-value is the **probability** (under H_0) of observing a value of the test statistic the same as or more extreme than what was actually observed

Hence, we calculate the p-value in this case as

$$v = P(X \geq k; n, p_0) = \sum_{i=k}^n \binom{n}{i} p_0^i (1 - p_0)^{n-i} = 1 - P(X < k; n, p_0)$$

Reject H_0 if $v < \alpha$

Example

A chip manufacturer claims that no more than 2% of its chips are defective.

A company has purchased 300 of these chips.

If 10 of these chips are defective, can the claim be rejected with 5% significance?

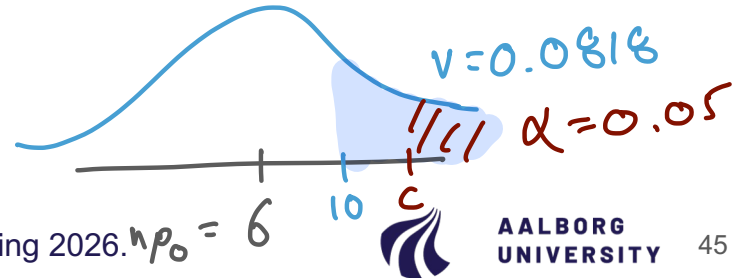
Solution

$H_0: p \leq p_0 = 0.02$, $n = 300$, and $k = 10$
We calculate the p-value as

$$\alpha = 0.05$$

$$Pr(X_{300} \geq 10; \eta, p_0) = \sum_{i=10}^{300} \binom{300}{i} (0.02)^i (1 - 0.02)^{300-i} = 0.0818 > 0.05$$

Since $v = 0.0818 > 0.05$, we cannot reject H_0



Solution with normal approximation

Central Limit Theorem:

For $X_1, X_2, \dots, X_n$ be i.i.d. RVs with mean μ and variance σ^2 if n is large:

$$\sum_{i=1}^n X_i \approx N(\underline{n\mu}, \underbrace{n\sigma^2}_{np})$$

$$\sigma^2 = p(1-p)$$

Hence, if n is large and the RVs are Bernoulli, we can use the Z-test with

$$T = \frac{\overset{\text{observation}}{k} - np_0}{\sqrt{np_0(1-p_0)}}$$

For the example $\underline{T = 1.1605} < \underline{Z_{0.05} = 1.6448}$, so **we cannot reject H_0**

Solving the coin toss example with normal approx.

We start with the null hypothesis $H_0: p = 1/2$

Toss the coin **a large number of times** n and count the number of heads X

Generate an estimate for p as

$$\hat{p}_n = \frac{1}{n} \sum_{i=1}^n X_i = \frac{X}{n} \approx N(\mu, \sigma^2/n)$$

We define our **test statistic** as $T = |\hat{p}_n - p_0| / \sqrt{p_0(1 - p_0)/n}$ where $p_0 = 1/2$

Rejection region: $R = \{\hat{p}_n: T > Z_{\alpha/2}\}$

It is concluded with α significance that the coin is not fair if $|\hat{p}_n - 1/2| > \frac{Z_{\alpha/2}}{2\sqrt{n}}$

So, we can say that a coin is not fair if $\hat{p}_n > \frac{Z_{\alpha/2}}{2\sqrt{n}} + 0.5$ or $\hat{p}_n < 0.5 - \frac{Z_{\alpha/2}}{2\sqrt{n}}$

Example: Student poll. To be solved during exercises

Students are asked to choose between two options

Evil lecturer has an initial hypothesis H_0 . Can it be rejected from the poll results?

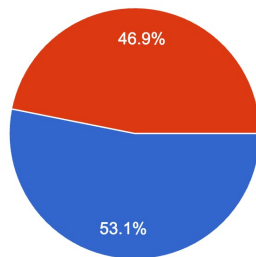
$X_i = 0$ if a student chooses option 1 and $X_i = 1$ if chooses option 2

Last year, 17 out of 32 students chose 1

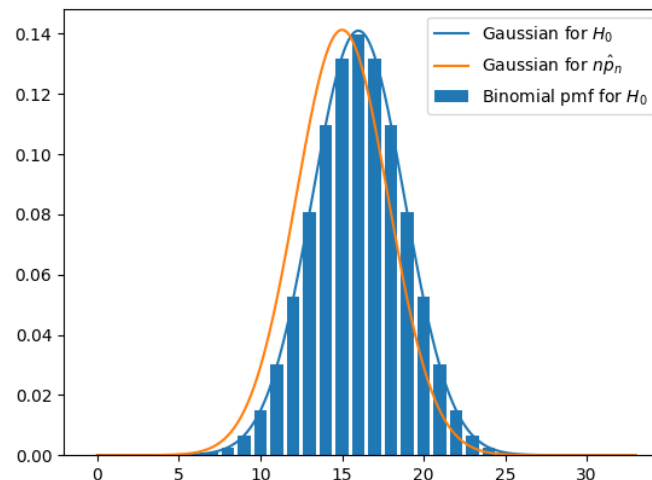
Can you reject H_0 : “students prefer 2”?

What do you prefer?

32 responses



- 1) Israel gives the rest of the theory on March 26. Students do the hypothesis testing exercises on their own. We do the workshop exercises on April 18.
- 2) Israel gives half of the theory on March 26 and the rest on April 18. We do the exercises for hypothesis testing and the workshop mixed with the lectures.



Syntax table

Parameter	Matlab syntax	Python syntax
$Z_{\alpha/2}$	<code>norminv(1-α/2)</code>	<code>scipy.stats.norm.ppf(1-α/2)</code>
$\Phi(T)$	<code>normcdf(T)</code>	<code>scipy.stats.norm.cdf(T)</code>

Summary

Summary

We have several types of tests based on:

- Number of populations involved, types of hypothesis, and assumptions

We need to choose the right one

If the distribution of the population is normal and the variance is known

- We can use the CIs with normal distribution for testing

Otherwise, we need slightly more advanced mathematics...